DevSecOps

Version Control

- Git
- GitLab
- GitHub
- Azure DevOps

Security

- JFrog Artifactory
- SonarQube

Build & Test Automation

- Jenkins
- GitLab
- GitHub Actions
- Travis CI
- TeamCity

Release Automation

- Docker
- OpenStack
- Azure
- AWS
- VMs

ML Models

- Logistic Regression
- SGD Classifier
- Complement NB
- Linear SVM
- Passive Aggressive
- Extra Trees
- Multinomial NB
- AdaBoost
- XGBoost
- Random Forest
- Ridge Classifier
- LightGBM
- Gradient Boosting

REST API

FastAPI

FastAPI

REST API

Quantum-Enhanced AI for NLP

- VQC
- QRC
- TensorHyper-VQC
- Quantum Kernel Methods (QSVM / QSVC)
- QCNN

Email Detection System Control UI

- C++
- Unreal Engine

Email Phishing Detection System Core

AI Models

- NLP (BERT, RoBERTa)
- LLMs (Mistrall-Small-24B-Insruct)
- LLM + Vision (Qwen3-VL-8B-Instruct)
- NLP (BERT, RoBERTa)
- LLM + Vision (Qwen3-VL-Embedding-8B)
- LLMs (Qwen3-4B)
- LLM + Vision (Qwen3-Embedding-8B)
- LLMs (DeepSeek-R1)
- LLM + Vision (Qwen3-VL-Reranker-8B)

REST API

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Sandbox Environment

- Behavioral instrumentation based on API hooking
- Capture of files created, modified and deleted during execution
- Network traffic capture in PCAP format
- Malware classification based on behavioral and network signatures
- Screenshots of the desktop taken during the execution of the malware
- Full memory dumps of the target system

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Semantic & Intent Analysis Module

- **ModernBERT**
- **all-mpnet-base-v2**
- **Qwen3-Embedding-0.6B**
- **cross-encoder/nli-deberta-v3-base**
- **SamLowe/roberta-base-go_emotions**
- **cardiffnlp/twitter-roberta-base-sentiment-latest**
- **vladinc/bigfive-regression-model**
- **MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli**
- **ntatiit/bert-banking77**
- **DeBERTa-v3-large-mnli**
- **dleemiller/finecat-nli-l**
- **GoEmotions-RoBERTa**

REST API

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Remote Forensics & Incident Response

- PowerForensics
- GRR Rapid Response
- Live-Forensicator
- osquery
- VelociRaptor
- HELK
- CAPEv2

FastAPI

- incoming email
- DOVECOT
- POSTFIX
- AMaViS
- SpamAssassin
- ClamAV
- Nginx
- mlmmj
- Roundcube
- Fail2ban
- DNSSEC,NSD4, SPF, TLS, DKIM, & DMARC
- munin
- OpenLDAP / ldapd
- Amavisd-new
- SOGo Groupware
- iRedAPD
- MTA-STS, SSHFP
- MySQL, MariaDB, PostgreSQL
- postgrey
- z-push
- Nextcloud
- duplicity

Email Detection System Backend

- NPM
- ElectronJS
- NodeJS

Email Detection System Frontend